

### Summary Statistics by Metro x Trip Category:

| Core City   | TripCategory | AvgDuration | StdDevDuration | MinDuration | Q1_Approx | Median_Approx | Q3_Approx | MaxDuration |
|-------------|--------------|-------------|----------------|-------------|-----------|---------------|-----------|-------------|
| Austin      | Commuter     | 13.15       | 54.65          | 1           | 5         | 7             | 10        | 1435        |
| Austin      | Leisure      | 30.44       | 72.43          | 0           | 7         | 14            | 30        | 1439        |
| Los Angeles | Commuter     | 14.18       | 56.14          | 1           | 5         | 8             | 13        | 1440        |
| Los Angeles | Leisure      | 39.95       | 132.63         | 0           | 7         | 14            | 28        | 1440        |
| Pittsburgh  | Commuter     | 27.12       | 112.47         | 1           | 7         | 11            | 17        | 1438        |
| Pittsburgh  | Leisure      | 65.96       | 172.47         | 0           | 10        | 22            | 55        | 1440        |
| Portland    | Commuter     | 14.31       | 32.29          | 1           | 6         | 9             | 15        | 1416        |
| Portland    | Leisure      | 25.69       | 45             | 1           | 7         | 14            | 30        | 1439        |

### Share of Trip Types (Short, Typical, Long) Within Each Metro and Category:

| MetroArea   | TripCategory | DurationBand                        | TripCount | PctWithinCategory |
|-------------|--------------|-------------------------------------|-----------|-------------------|
| Austin      | Commuter     | Shorter Trips (Bottom 25% of Metro) | 86327     | 46.57             |
| Austin      | Commuter     | Typical Trips (Middle 50% of Metro) | 90394     | 48.76             |
| Austin      | Commuter     | Longer Trips (Top 25% of Metro)     | 8660      | 4.67              |
| Austin      | Leisure      | Shorter Trips (Bottom 25% of Metro) | 262283    | 24.32             |
| Austin      | Leisure      | Typical Trips (Middle 50% of Metro) | 514682    | 47.72             |
| Austin      | Leisure      | Longer Trips (Top 25% of Metro)     | 301482    | 27.96             |
| Los Angeles | Commuter     | Shorter Trips (Bottom 25% of Metro) | 94451     | 38.76             |

|             |          |                                     |        |       |
|-------------|----------|-------------------------------------|--------|-------|
| Los Angeles | Commuter | Typical Trips (Middle 50% of Metro) | 132953 | 54.56 |
| Los Angeles | Commuter | Longer Trips (Top 25% of Metro)     | 16280  | 6.68  |
| Los Angeles | Leisure  | Shorter Trips (Bottom 25% of Metro) | 209596 | 24.42 |
| Los Angeles | Leisure  | Typical Trips (Middle 50% of Metro) | 391614 | 45.62 |
| Los Angeles | Leisure  | Longer Trips (Top 25% of Metro)     | 257174 | 29.96 |
| Pittsburgh  | Commuter | Shorter Trips (Bottom 25% of Metro) | 21611  | 42.18 |
| Pittsburgh  | Commuter | Typical Trips (Middle 50% of Metro) | 27756  | 54.17 |
| Pittsburgh  | Commuter | Longer Trips (Top 25% of Metro)     | 1867   | 3.64  |
| Pittsburgh  | Leisure  | Shorter Trips (Bottom 25% of Metro) | 104435 | 24.04 |
| Pittsburgh  | Leisure  | Typical Trips (Middle 50% of Metro) | 211829 | 48.76 |
| Pittsburgh  | Leisure  | Longer Trips (Top 25% of Metro)     | 118143 | 27.2  |
| Portland    | Commuter | Shorter Trips (Bottom 25% of Metro) | 62177  | 39.49 |
| Portland    | Commuter | Typical Trips (Middle 50% of Metro) | 82899  | 52.65 |
| Portland    | Commuter | Longer Trips (Top 25% of Metro)     | 12368  | 7.86  |
| Portland    | Leisure  | Shorter Trips (Bottom 25% of Metro) | 302344 | 26.82 |

#### **Dockless vs. Trip Duration by Metro & Category:**

| MetroArea | TripCategory | TripMode           | TripCount | AvgDuration_Min | StdDevDuration_Min |
|-----------|--------------|--------------------|-----------|-----------------|--------------------|
| Austin    | Commuter     | Station-to-Station | 177446    | 13.15           | 54.65              |
| Austin    | Leisure      | Station-to-Station | 1083698   | 30.44           | 72.43              |

|             |          |                    |        |       |        |
|-------------|----------|--------------------|--------|-------|--------|
| Los Angeles | Commuter | Station-to-Station | 229552 | 14.18 | 56.14  |
| Los Angeles | Leisure  | Station-to-Station | 834888 | 39.95 | 132.63 |
| Pittsburgh  | Commuter | Dockless Involved  | 4525   | 23.12 | 85.36  |
| Pittsburgh  | Commuter | Station-to-Station | 45804  | 27.51 | 114.8  |
| Pittsburgh  | Leisure  | Dockless Involved  | 32351  | 56.95 | 156.93 |
| Pittsburgh  | Leisure  | Station-to-Station | 399947 | 66.69 | 173.64 |
| Portland    | Commuter | Dockless Involved  | 59586  | 16.63 | 43.94  |
| Portland    | Commuter | Station-to-Station | 95010  | 12.86 | 21.92  |
| Portland    | Leisure  | Dockless Involved  | 514141 | 23.88 | 53.8   |
| Portland    | Leisure  | Station-to-Station | 615851 | 27.2  | 35.98  |

#### Average Trip Distance by Metro and Category:

| MetroArea   | TripCategory | Trips  | AvgTripDistance_Miles | StdDevTripDistance_Miles |
|-------------|--------------|--------|-----------------------|--------------------------|
| Austin      | Commuter     | 177089 | 0.7                   | 0.42                     |
| Austin      | Leisure      | 882690 | 0.81                  | 0.5                      |
| Los Angeles | Commuter     | 219258 | 0.8                   | 0.52                     |
| Los Angeles | Leisure      | 617019 | 0.85                  | 0.7                      |
| Pittsburgh  | Commuter     | 45168  | 1.18                  | 0.78                     |
| Pittsburgh  | Leisure      | 272532 | 1.19                  | 0.87                     |
| Portland    | Commuter     | 95010  | 0.64                  | 0.5                      |
| Portland    | Leisure      | 516454 | 0.56                  | 0.46                     |

#### Distance + Duration Efficiency:

| MetroArea | TripCategory | AvgTripDistance_Miles | AvgTripDuration_Min | AvgMilesPerMinute |
|-----------|--------------|-----------------------|---------------------|-------------------|
| Austin    | Commuter     | 0.7                   | 13.09               | 0.0914            |
| Austin    | Leisure      | 0.81                  | 26.66               | 0.0643            |

|             |          |      |       |        |
|-------------|----------|------|-------|--------|
| Los Angeles | Commuter | 0.8  | 12.35 | 0.0902 |
| Los Angeles | Leisure  | 0.85 | 26.15 | 0.0708 |
| Pittsburgh  | Commuter | 1.18 | 27.46 | 0.0975 |
| Pittsburgh  | Leisure  | 1.19 | 52.47 | 0.0731 |
| Portland    | Commuter | 0.64 | 12.86 | 0.0702 |
| Portland    | Leisure  | 0.56 | 23.16 | 0.0437 |

## SQL Code:

```
DROP TEMPORARY TABLE IF EXISTS ClassifiedTrips;
```

```
CREATE TEMPORARY TABLE ClassifiedTrips AS
SELECT
    TripID,
    MetroID,
    StartDate,
    StartTime,
    StartDayOfWeek,
    FromStationID,
    ToStationID,
    UserType,
    TripDuration_Minutes,
    CASE
        WHEN StartDayOfWeek BETWEEN 1 AND 5
            AND UserType = 'Subscriber'
            AND (
                (StartTime BETWEEN '06:30:00' AND '09:30:00')
                OR (StartTime BETWEEN '16:00:00' AND '19:00:00')
            )
            AND FromStationID <> ToStationID
        THEN 'Commuter'
        ELSE 'Leisure'
    END AS TripCategory
FROM Rentals
WHERE NOT (
    TripDuration_Minutes = 0
    AND FromStationID <> ToStationID
);
```

--Compute Quartiles into a Temporary Table

```
DROP TEMPORARY TABLE IF EXISTS TripQuartiles;
```

```
CREATE TEMPORARY TABLE TripQuartiles AS
WITH RankedDurations AS (
    SELECT
        MetroID,
        TripCategory,
        TripDuration_Minutes,
        PERCENT_RANK() OVER (
            PARTITION BY MetroID, TripCategory
            ORDER BY TripDuration_Minutes
```

```

        ) AS pr
    FROM ClassifiedTrips
)
SELECT
    MetroID,
    TripCategory,
    MAX(CASE WHEN pr <= 0.25 THEN TripDuration_Minutes END) AS Q1,
    MAX(CASE WHEN pr <= 0.75 THEN TripDuration_Minutes END) AS Q3
FROM RankedDurations
GROUP BY MetroID, TripCategory;

```

--Create Clean Trips (without outliers) Using That Table  
DROP TEMPORARY TABLE IF EXISTS ClassifiedTrips\_Clean;

```

CREATE TEMPORARY TABLE ClassifiedTrips_Clean AS
SELECT C.*
FROM ClassifiedTrips C
JOIN TripQuartiles Q
    ON C.MetroID = Q.MetroID
    AND C.TripCategory = Q.TripCategory
WHERE C.TripDuration_Minutes BETWEEN
    (Q.Q1 - 1.5 * (Q.Q3 - Q.Q1))
    AND (Q.Q3 + 1.5 * (Q.Q3 - Q.Q1));

```

--Diagnostic Check for How Many Outliers Removed

```

SELECT
    M.CoreCity,
    C.TripCategory,
    COUNT(*) AS OriginalTrips,
    SUM(CASE WHEN CL.TripID IS NULL THEN 1 ELSE 0 END) AS RemovedAsOutlier,
    ROUND(
        100.0 * SUM(CASE WHEN CL.TripID IS NULL THEN 1 ELSE 0 END) / COUNT(*),
        2
    ) AS PctRemoved
FROM ClassifiedTrips C
JOIN MetroArea M ON C.MetroID = M.MetroID
LEFT JOIN ClassifiedTrips_Clean CL
    ON C.TripID = CL.TripID
GROUP BY M.CoreCity, C.TripCategory;

```

--1) Retrive Table With All Trips to Create Box Plot

```

SELECT
    C.TripID,
    M.CoreCity,

```

```
    C.TripDuration_Minutes,  
    C.TripCategory  
FROM ClassifiedTrips C  
JOIN MetroArea M ON C.MetroID = M.MetroID;
```

-- 2) Summary Statistics by Metro x Trip Category

```
DROP TEMPORARY TABLE IF EXISTS ClassifiedTrips_Ranked;
```

```
CREATE TEMPORARY TABLE ClassifiedTrips_Ranked AS  
SELECT  
    TripID,  
    MetroID,  
    TripCategory,  
    TripDuration_Minutes,  
    PERCENT_RANK() OVER (  
        PARTITION BY MetroID, TripCategory  
        ORDER BY TripDuration_Minutes  
    ) AS pr  
FROM ClassifiedTrips;
```

```
WITH Statistics_Clean AS (  
    SELECT  
        MetroID,  
        TripCategory,  
        MIN(TripDuration_Minutes) AS MinDuration,  
        MAX(CASE WHEN pr <= 0.25 THEN TripDuration_Minutes END) AS Q1_Approx,  
        MAX(CASE WHEN pr <= 0.50 THEN TripDuration_Minutes END) AS Median_Approx,  
        MAX(CASE WHEN pr <= 0.75 THEN TripDuration_Minutes END) AS Q3_Approx,  
        MAX(TripDuration_Minutes) AS MaxDuration  
    FROM ClassifiedTrips_Ranked  
    GROUP BY MetroID, TripCategory  
)  
SELECT  
    M.CoreCity,  
    C.TripCategory,  
    ROUND(AVG(C.TripDuration_Minutes), 2) AS AvgDuration,  
    ROUND(STDDEV(C.TripDuration_Minutes), 2) AS StdDevDuration,  
    S.MinDuration,  
    S.Q1_Approx,  
    S.Median_Approx,  
    S.Q3_Approx,  
    S.MaxDuration  
FROM ClassifiedTrips_Clean AS C
```

```

JOIN MetroArea M
  ON C.MetroID = M.MetroID
JOIN Statistics_Clean S
  ON C.MetroID = S.MetroID
  AND C.TripCategory = S.TripCategory
GROUP BY
  M.CoreCity,
  C.TripCategory
ORDER BY M.CoreCity, C.TripCategory;

```

--3) Share of Trip Types (Short, Typical, Long) Within Each Metro and Category

```

WITH MetroRankedDurations AS (
  SELECT
    MetroID,
    TripID,
    TripCategory,
    TripDuration_Minutes,
    PERCENT_RANK() OVER (
      PARTITION BY MetroID
      ORDER BY TripDuration_Minutes
    ) AS pr
  FROM ClassifiedTrips_Clean
)
SELECT
  M.CoreCity AS MetroArea,
  R.TripCategory,
  CASE
    WHEN R.pr <= 0.25 THEN 'Shorter Trips (Bottom 25% of Metro)'
    WHEN R.pr <= 0.75 THEN 'Typical Trips (Middle 50% of Metro)'
    ELSE 'Longer Trips (Top 25% of Metro)'
  END AS DurationBand,
  COUNT(*) AS TripCount,
  ROUND(
    100.0 * COUNT(*) /
    SUM(COUNT(*)) OVER (PARTITION BY M.CoreCity, R.TripCategory),
    2
  ) AS PctWithinCategory
FROM MetroRankedDurations R
JOIN MetroArea M ON R.MetroID = M.MetroID
GROUP BY M.CoreCity, R.TripCategory, DurationBand
ORDER BY M.CoreCity, R.TripCategory,
  FIELD(DurationBand,
    'Shorter Trips (Bottom 25% of Metro)',
    'Typical Trips (Middle 50% of Metro)',

```



'Longer Trips (Top 25% of Metro)');

--4) Rentals beginning Dockless (not at a station) vs. Trip Duration

```
SELECT
  M.CoreCity AS MetroArea,
  C.TripCategory,
  CASE
    WHEN C.FromStationID = -5000 OR C.ToStationID = -5000 THEN 'Dockless Involved'
    ELSE 'Station-to-Station'
  END AS TripMode,
  COUNT(*) AS TripCount,
  ROUND(AVG(C.TripDuration_Minutes), 2) AS AvgDuration_Min,
  ROUND(STDDEV(C.TripDuration_Minutes), 2) AS StdDevDuration_Min
FROM ClassifiedTrips_Clean C
JOIN MetroArea M ON C.MetroID = M.MetroID
GROUP BY M.CoreCity, C.TripCategory, TripMode
ORDER BY M.CoreCity, C.TripCategory, TripMode;
```

--Base Table for Distance Analysis

DROP TEMPORARY TABLE IF EXISTS SpatialTrips;

CREATE TEMPORARY TABLE SpatialTrips AS

```
SELECT
  C.TripID,
  C.MetroID,
  C.TripCategory,
  C.TripDuration_Minutes,

  FS.StationName AS FromStationName,
  TS.StationName AS ToStationName,

  FS.Latitude AS FromLat,
  FS.Longitude AS FromLon,
  TS.Latitude AS ToLat,
  TS.Longitude AS ToLon,

  -- Haversine formula (miles)
  3959 * 2 * ASIN(
    SQRT(
      POWER(SIN(RADIANS(TS.Latitude - FS.Latitude) / 2), 2) +
      COS(RADIANS(FS.Latitude)) * COS(RADIANS(TS.Latitude)) *
      POWER(SIN(RADIANS(TS.Longitude - FS.Longitude) / 2), 2)
    )
  ) AS TripDistance_Miles
```

```

FROM ClassifiedTrips_Clean C
JOIN Stations FS
  ON C.FromStationID = FS.StationID
  AND C.MetroID = FS.MetroID
JOIN Stations TS
  ON C.ToStationID = TS.StationID
  AND C.MetroID = TS.MetroID

```

-- Exclude dockless trips (no real station location)

```

WHERE
  C.FromStationID <> -5000
  AND C.ToStationID <> -5000
  AND FS.Latitude > 0
  AND FS.Longitude < 0
  AND TS.Latitude > 0
  AND TS.Longitude < 0;

```

-- 5) Avg Trip Distance by Category × Metro

```

SELECT
  M.CoreCity AS MetroArea,
  S.TripCategory,
  COUNT(*) AS Trips,
  ROUND(AVG(S.TripDistance_Miles), 2) AS AvgTripDistance_Miles,
  ROUND(STDDEV(S.TripDistance_Miles), 2) AS StdDevTripDistance_Miles
FROM SpatialTrips S
JOIN MetroArea M ON S.MetroID = M.MetroID
GROUP BY M.CoreCity, S.TripCategory
ORDER BY M.CoreCity, S.TripCategory;

```

-- 6) Distance + Duration Efficiency (Miles per Minute)

```

SELECT
  M.CoreCity AS MetroArea,
  S.TripCategory,
  ROUND(AVG(S.TripDistance_Miles), 2) AS AvgTripDistance_Miles,
  ROUND(AVG(S.TripDuration_Minutes), 2) AS AvgTripDuration_Min,
  ROUND(
    AVG(S.TripDistance_Miles / NULLIF(S.TripDuration_Minutes, 0)),
    4
  ) AS AvgMilesPerMinute
FROM SpatialTrips S
JOIN MetroArea M ON S.MetroID = M.MetroID
GROUP BY M.CoreCity, S.TripCategory
ORDER BY M.CoreCity, S.TripCategory;

```

